

Towards A New Era of Geo-Foundation Models: Expert-Guided Multimodal Alignment and Geospatial Context Awareness

Ting Han^{1,2}, Huan Chen¹, Chaolei Wang¹, Yilan Ren¹, Meiliu Wu^{2,*}

¹ School of Geospatial Engineering and Science, Sun Yat-sen University, Zhuhai, China

² School of Geographical and Earth Sciences, University of Glasgow, Glasgow, United Kingdom

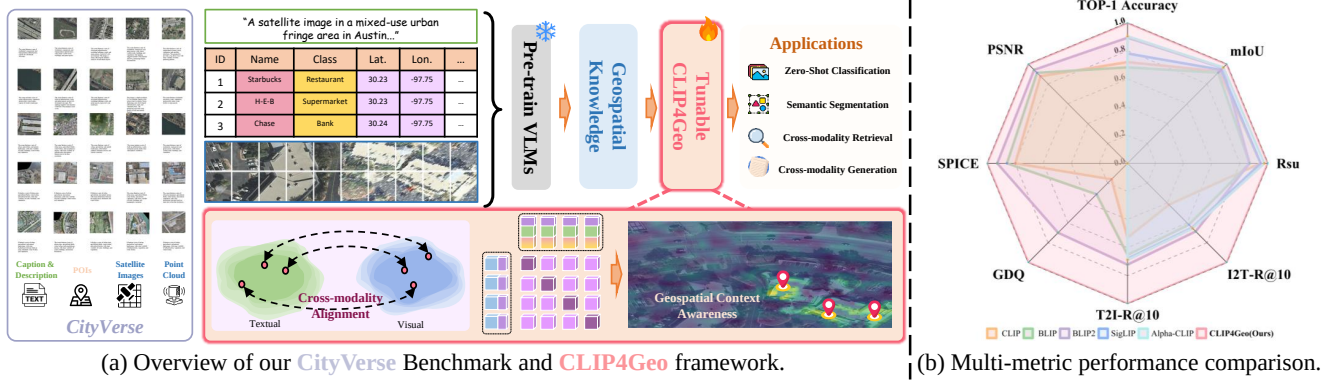


Figure 1: Our proposed framework for developing Geo-Foundation Models. (a) CLIP4Geo (Contrastive Language–Image Pretraining enhanced by Geospatial intelligence) is fine-tuned on our CityVerse benchmark, integrating multimodal geospatial data into a unified representation to better support various geospatial tasks. (b) Multi-metric performance comparisons show that CLIP4Geo consistently outperforms state-of-the-art baselines on various tasks, highlighting the effectiveness of expert-guided multimodal alignment and geospatial context awareness in improving GeoFMs’ capabilities.

Abstract

Foundation Models (FMs) have demonstrated significant potential for geospatial analysis. As geospatial big data (e.g., geo-tagged text and images) continues to grow, FMs have evolved into geospatial foundation models (GeoFMs), the FMs capable of geospatial reasoning. However, two critical tasks remain challenging: (1) *how to align various geospatial modalities for model training*, and (2) *how to enhance FMs’ geospatial context awareness*. To fill these gaps, this work develops **CLIP4Geo**, a unified GeoFM that integrates satellite imagery, LiDAR point clouds, geo-tagged points of interest (POIs), and textual descriptions into a shared representation space. Specifically, to empower the model with geospatial intelligence, we design a **Geospatial Expert Knowledge System** that tailors both textual and visual information and encodes them into shared geospatial embeddings. Furthermore, we propose a novel **Geospatial Context Awareness Transformer** to guide the cross-attention mechanism, which leverages geospatial context position embedding to effectively capture spatial dependencies across modalities. We also introduce **CityVerse**, a large-scale, spatially aligned, and multimodal benchmark dataset covering diverse urban

regions in the U.S. and China, supporting a range of geospatial downstream tasks (e.g., zero-shot classification, semantic segmentation / localization, and cross-modality retrieval and generation). Extensive experiments demonstrate that CLIP4Geo consistently outperforms state-of-the-art FM baselines, highlighting the importance of expert-guided multimodal alignment and geospatial context embeddings. Our work provides a scalable framework and comprehensive benchmark to facilitate the future development of GeoFMs for various geospatial applications. Code and dataset are available at <https://github.com/Ting-Devin-Han/CLIP4Geo.git>.

CCS Concepts

• Computing methodologies → Scene understanding.

Keywords

CLIP4Geo, Geospatial Foundation Model, Multimodal Learning, CityVerse, Geospatial Benchmark Dataset, Geospatial AI, Geographic Information Science, Remote Sensing

ACM Reference Format:

Ting Han^{1,2}, Huan Chen¹, Chaolei Wang¹, Yilan Ren¹, Meiliu Wu^{2,*}. 2025. Towards A New Era of Geo-Foundation Models: Expert-Guided Multimodal Alignment and Geospatial Context Awareness. In *The 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25)*, November 3–6, 2025, Minneapolis, MN, USA. ACM, New York, NY, USA, 12 pages. <https://doi.org/10.1145/3748636.3762756>



This work is licensed under a Creative Commons Attribution 4.0 International License. [SIGSPATIAL '25](https://creativecommons.org/licenses/by/4.0/), Minneapolis, MN, USA

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2086-4/2025/11

<https://doi.org/10.1145/3748636.3762756>

1 Introduction

In recent years, deep learning-based artificial intelligence (AI) [4, 5, 29, 34, 41] has become a powerful tool for geospatial analysis, contributing to more efficient urban planning [6], enhanced public service quality [50], and the advancement of smart cities [20, 48]. In particular, the recent rise of foundation models (FMs) [38] has profoundly reshaped the research paradigm of geospatial information science and remote sensing [14, 60]. These models, pre-trained on large-scale datasets using self-supervised learning methods, have demonstrated remarkable capabilities in reasoning, generalization, and emergent behavior [1, 22]. Furthermore, they can be subsequently fine-tuned for various geospatial downstream tasks [16, 64], ushering in a new era of geo-foundation models (GeoFMs).

GeoFMs [1, 19] are designed for integrating information from diverse geo-tagged sources, uncovering the spatiotemporal dynamics in various scenarios, and enabling a deeper understanding of spatial relationships across data modalities and stronger geospatial reasoning capabilities. For example, SatMAE [11] employs Masked Autoencoder (MAE) [17] to learn generalizable visual representations from multi-source satellite imagery via self-supervised pre-training framework. RemoteCLIP [28] adopts a CLIP-based paradigm [27, 39] to align remote sensing imagery with textual description, significantly improving semantic analysis. Moreover, models such as ReFound [48], UrbanVLM [16] and GeoGPT [61] aim to integrate multi-modalities to support various geospatial tasks, exhibiting strong generalization capabilities and representing a new direction for knowledge transfer in geospatial intelligence.

However, in multimodal settings, textual description of imagery often provide generic semantic annotations and fail to incorporate structured geospatial context (e.g., coordinates, administrative boundaries, or their spatial relationships), causing most GeoFMs to overlook the spatial consistency and dependencies inherent in geographic data (Fig. 2). In addition, although some GeoFMs (e.g., GeoCLIP [43] and PDFM [1]) explore the alignment between geo-locations and images, yet relying solely on GPS coordinates is insufficient for a comprehensive understanding of imagery, as it fails to capture the geospatial context information of the given area. That said, **the effective alignment of geospatial context across different data modalities and the construction of a unified geospatial representation remain challenges**. Moreover, the lack of large-scale, high-quality, multimodal geospatial datasets (e.g., integrating images, text, coordinates, attributes and 3D information) continues to impede the training and evaluation of comprehensive GeoFMs. Addressing these limitations is critical to unlocking the full potential of GeoFMs, and these observations motivate us to explore several key research questions (RQs):

- RQ1: *How effective is multimodal geospatial context for improving FMs' geospatial reasoning capabilities?*
- RQ2: *How can different geospatial modalities be effectively aligned and integrated to build a unified representation for downstream tasks?*
- RQ3: *How can we construct a comprehensive benchmark that is representative for diverse modalities and scenarios to support real-world geospatial applications?*

Our work aims to answer these questions by developing **CLIP4Geo**, a unified GeoFM that aligns satellite images, LiDAR point clouds,

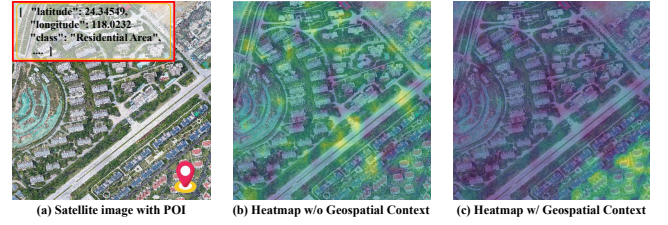


Figure 2: The effectiveness of geospatial context awareness and alignment. (b) The attention maps without geospatial context are diffused and lack semantic focus. (c) In contrast, incorporating geospatial context awareness leads to sharper and more concentrated responses in semantically meaningful regions such as POI-located areas.

geo-tagged POIs, and textual description into a shared representation space. We also construct **CityVerse**, a large-scale, multimodal benchmark specifically curated to evaluate geospatial alignment and task generalization across diverse contexts. To demonstrate the effectiveness of our framework (Fig. 1), extensive experiments are conducted on four downstream tasks: zero-shot classification, semantic segmentation / localization, cross-modality retrieval, and cross-modality generation. As results, **CLIP4Geo consistently outperforms existing GeoFMs**, highlighting the importance of multimodal fusion and geospatial context. Our main contributions are summarized as follows:

- We develop a novel GeoFM that explicitly incorporates geospatial context from both visual and textual inputs, enabling spatially consistent and semantically grounded understanding across diverse scenarios.
- We design a unified architecture that aligns satellite imagery, geo-tagged texts, POIs and LiDAR point clouds into a shared representation space, enhancing cross-modality geospatial understanding and interpretability.
- We construct CityVerse, a large-scale, spatially aligned, and semantically diverse benchmark dataset that supports training and evaluation across multiple geospatial tasks, addressing the lack of benchmark data.
- We conduct extensive experiments on different downstream tasks, demonstrating the effectiveness and extensibility of our framework, and laying the foundation for future research in GeoFMs.

2 Related Work

2.1 Vision-Language Foundation Models

Vision-Language Foundation Models (VLMs; e.g., CLIP [33, 39] and BLIP [25, 26]) have made remarkable progress by learning joint representations of images and text from large-scale datasets. These models are pre-trained based on contrastive [44] or generative objectives [12], enabling strong zero-shot capabilities across diverse vision-language tasks. These models have shown effectiveness in classification [47], retrieval [30], and captioning [33], thanks to their ability to capture high-level semantic alignment between visual and textual modalities. Recent extensions further explore multi-stage

learning and multimodal reasoning to support visual Q&A and retrieval tasks. [32, 60].

However, most existing VLMs are trained based on general text and imagery and often lack geospatial-specific information, limiting their effectiveness in geospatial domains (e.g., remote sensing reasoning). In this work, we address these limitations by **explicitly encoding tailored, geospatial image-text pairs into visual and textual embeddings to enhance the model’s geospatial understanding**.

2.2 Geospatial Foundation Models

Recent advances in GeoFMs [16] aim to bring the strengths of general-purpose VLMs into geospatial domains [48]. These models are typically pre-trained on geospatial imagery (e.g., satellite images and street view images) paired with captions or metadata, enabling them to support geospatial tasks such as land use classification [21], scene captioning [63], and geo-localization [46]. Some approaches (GeoCLIP [43], RemoteCLIP [28], LRSCLIP [8]) leverage large-scale imagery datasets to adapt CLIP-based architectures, while others explore the fusion of modality-specific encoders (e.g., incorporating a location encoder [23], text encoder and image encoder) to capture spatial patterns and geospatial context.

Despite promising progress, current GeoFMs still face limitations. First, most models focus on image-text alignment but neglect the incorporation of structured geospatial information (e.g., geo-locations and spatial relationships of the input). Second, integrating different geospatial modalities presents unique challenges due to varying geographic scales and semantic granularity. Moreover, these models’ capacity for geospatial reasoning remains constrained due to a lack of effective geographic context modeling.

To address these limitations, our work introduces a unified framework with expert-guided geospatial knowledge, which leverages the architectural strengths of MAE [17] and CLIP [39]. Specifically, this framework first utilizes MAE to learn robust visual representations by linking textual description with the paired masked image. Additionally, **the framework integrates satellite images, LiDAR, geo-tagged POIs, and geospatial-context text description into a shared representation space, enabling more precise geospatial consistency and cross-modality alignment, as well as richer semantic understanding with a focus on different aspects of geospatial context** (e.g., land use interpretation and spatial pattern description).

2.3 Geospatial Benchmarks and Datasets

The development of GeoFMs heavily relies on high-quality, large-scale datasets [7, 13, 24, 52] that capture the diverse characteristics of geospatial context. Existing datasets (e.g., EarthNets [49], EuroSAT [18], and RESISC45 [10]) provide labeled satellite images for classification tasks, while others (e.g., RSICD [30] and RSITMD [55]) support image-text retrieval and captioning. These benchmarks have facilitated the progress of GeoFMs by providing standardized evaluation protocols across various tasks.

However, most current datasets focus on a single or dual modality (e.g., image-text or image-label) and lack multimodal alignment. Additionally, there is a scarcity of datasets for extracting comprehensive geospatial context, limiting the capacity of models to perform

Benchmark	Tasks	Modality
EarthNets [49]	Class., SemSeg., OD.	MSI
SustainBench [52]	Class., SemSeg., Regr.	MSI, StreetView
GEO-Bench [24]	Class., SemSeg.	MSI, SAR ¹ , DSM ²
PhilEO Bench [13]	SemSeg.	MSI
FoMO Bench [7]	Class., SemSeg., OD.	MSI
RS5M [62]	Class., SemSeg., Retri.	SI, Text
LRS2M [8]	Class., SemSeg., Retri.	SI, Text
<i>CityVerse</i>	Class., SemSeg., Retri., Gen.	SI, PC, POI, Text

¹ SAR: Synthetic Aperture Radar.

² DSM: Digital Surface Model.

Table 1: Comparison of the *CityVerse* Benchmark with existing geospatial vision benchmarks in task diversity and modality coverage. The tasks include zero-shot classification (Class.), semantic segmentation (SemSeg.), object detection (OD.), regression (Regr.), retrieval (Retri.), and generation (Gen.) The MSI and SI denote the Multispectral Imagery and Satellite Imagery.

fine-grained geospatial reasoning. In contrast, we introduce a new multimodal geospatial benchmark, CityVerse (Tab. 1), which **integrates satellite imagery, LiDAR point clouds, geo-tagged POIs, and tailored textual metadata focused on different aspects of geospatial context. It facilitates the training and evaluation of GeoFMs by diversifying feasible tasks and modalities.** More details can be found in Section 5.

3 Preliminaries

3.1 Formulation

We partition the entire study area into B blocks. Each block is associated with a collection of different geospatial data modalities: satellite images $\mathcal{I}$, LiDAR point clouds $\mathcal{L}$, POIs $\mathcal{P}$, and geospatial-context description $\mathcal{T}$.

Satellite Imagery. Satellite images $\mathcal{I} \in \mathbb{R}^{H \times W \times 3}$ provide an overhead view of each block, where H and W are the image height and width, respectively.

LiDAR Point Clouds. The LiDAR point clouds $\mathcal{L} \in \mathbb{R}^{N \times 6}$ contain fine-grained 3D information. Each point consists of 3D coordinates (x, y, z) and RGB color values (r, g, b) . N denotes the total number of points per block.

Points of Interest (POIs). The POI data $\mathcal{P}$ are represented as structured tuples $p = (x, y, c)$, where (x, y) are geographic coordinates and c is the categorical label indicating semantic type (e.g., commercial, residential).

Textual Description. The textual description $\mathcal{T}$ includes automatically generated and manually curated sentences and captions describing different aspects of geospatial context in each block (e.g., POI’s land-use tag and spatial layout of identified objects).

3.2 Tasks Definition

Zero-shot Classification. This task targets scene-level categorization without labeled examples for the target classes. The fused multimodal embedding $\mathbf{z}_b$ for block b is the input, and the predicted

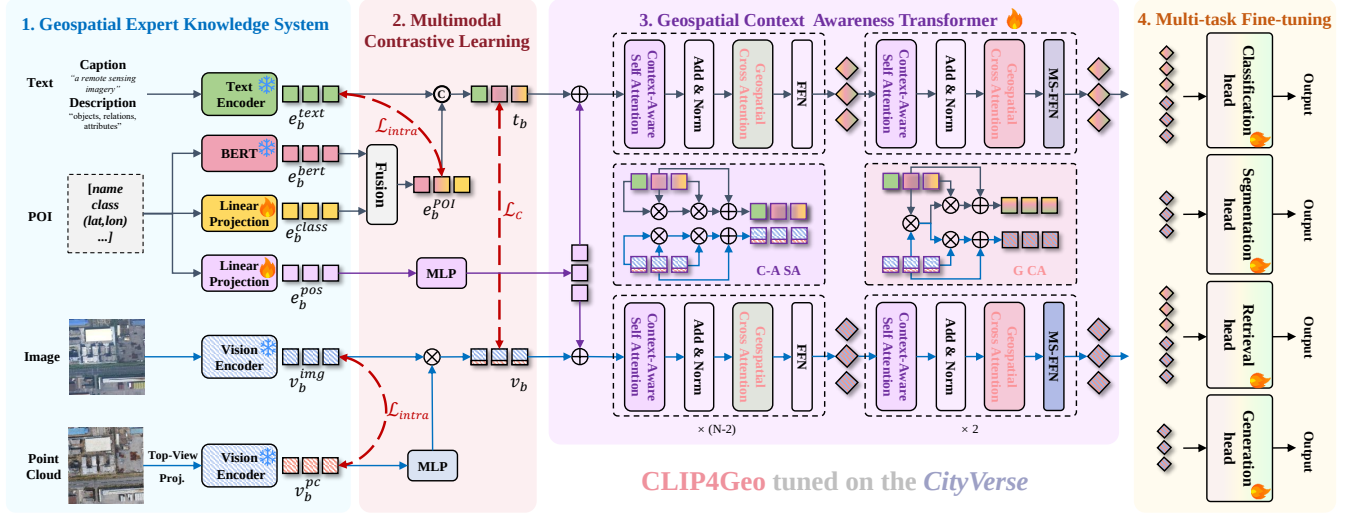


Figure 3: Overview of the proposed **CLIP4Geo** framework. It consists of three main components: (1) Geospatial Expert Knowledge System, which encodes textual description, geo-tagged POIs, as well as visual-based satellite images and point clouds; (2) Multimodal Contrastive Learning aligns textual and visual feature embeddings via contrastive objectives; (3) Geospatial Context Awareness Transformer models spatial dependencies using context-aware self-attention and geospatial cross-attention layers; and (4) Multi-task Fine-tuning supports four geospatial tasks - zero-shot classification, segmentation / localization, cross-modality retrieval, and cross-modality generation, based on **CLIP4Geo** tuned on the proposed *CityVerse* benchmark.

class label $\hat{y}_b \in \{1, \dots, K\}$ is the output. With textual description of categories $\{c_1, \dots, c_K\}$ embedded into the shared space, prediction is performed by similarity:

$$\hat{y}_b = \arg \max_k \cos(z_b, c_k). \quad (1)$$

Semantic Segmentation / Localization. This task performs dense prediction to assign a semantic label to each pixel in the image, supporting both fine-grained object segmentation and localization. The satellite image I_b and fused features are the input and the semantic masks $S_b \in \{1, \dots, C\}^{H \times W}$ are the output. Training uses pixel-wise cross-entropy loss:

$$\mathcal{L}_{\text{seg}} = - \sum_{i,j,c} y_{i,j,c} \log(\hat{y}_{i,j,c}). \quad (2)$$

Cross-modality Retrieval. This task evaluates the model's ability to align visual and textual representations by retrieving the most relevant images/text from a database based on a textual/visual query. For text-to-image retrieval, the query embedding $q_{\text{text}} \in \mathbb{R}^d$ is the input, and the output is a ranked list of image embeddings $\{v_1, \dots, v_N\}$, sorted by their cosine similarity:

$$\text{Top-}k_{\text{image}} = \text{argsort}_{i=1}^N \cos(q_{\text{text}}, v_i). \quad (3)$$

For image-to-text retrieval, the input is a visual query embedding $q_{\text{img}} \in \mathbb{R}^d$ extracted from satellite imagery or point clouds, and the output is a ranked list of textual embeddings $\{t_1, \dots, t_M\}$:

$$\text{Top-}k_{\text{text}} = \text{argsort}_{j=1}^M \cos(q_{\text{img}}, t_j). \quad (4)$$

Cross-modality Generation. This generative task requires the model to produce a coherent natural language description of a multimodal scene. This task takes the fused latent representation z from image, point cloud, POI, and text as input, then outputs the

caption sequence $\hat{t} = (w_1, \dots, w_T)$ of length T . The model is trained to maximize the log-likelihood of the ground-truth caption:

$$\mathcal{L}_{\text{gen}} = - \sum_{t=1}^T \log P(w_t | w_{<t}, z). \quad (5)$$

In addition to caption generation, we include a masked image reconstruction task to enhance structural understanding and multimodal fusion. The partially masked image $\tilde{I}$ and fused representation z are used to reconstruct image $\hat{I}$. The model learns to recover masked pixels using a decoder and is optimized with the mean squared error loss:

$$\mathcal{L}_{\text{mask}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} \|x_i - \hat{x}_i\|^2, \quad (6)$$

where $\mathcal{M}$ is the set of masked pixel indices, x_i is the original pixel value, and $\hat{x}_i$ is the reconstructed RGB value.

4 CLIP4Geo Framework

In this section, we detail our CLIP4Geo framework (Fig. 3). Our CLIP4Geo consists of three main components: (1) Geospatial Expert Knowledge System, (2) Multimodal Contrastive Learning, and (3) Geospatial Context Awareness Transformer, with subsequent Multi-task Fine-tuning for model evaluation.

4.1 Geospatial Expert Knowledge System

In this study, the Geospatial Expert Knowledge System is designed to transform geo-locations, geographic entities, and their geospatial context into enriched multimodal embeddings based on expert-guided extraction of both textual and visual inputs.

4.1.1 Textual Knowledge Extraction. Textual Knowledge Extraction aims to encode natural language scene description and POI information into geospatial semantic embeddings.

For each block b , we first organize the POI set $\mathcal{P}_b = \{p_1, \dots, p_m\}$ into a pseudo-sentence (including the name, address, semantic type, etc.) to ensure compatibility with the BERT tokenizer. This sentence is then passed through a pre-trained BERT model to obtain contextualized token embeddings $\mathbf{e}_b^{\text{BERT}}$, capturing the semantic relationships among POIs. To incorporate spatial information, we convert each POI's 2D geographic coordinates (x_i, y_i) into a 1D spatial index s_i using a Z-order space-filling curve, which preserves spatial locality. This 1D spatial index is further transformed into a dense position embedding $\mathbf{e}_i^{\text{pos}}$ via a multi-layer perceptron (MLP), allowing spatial proximity to be encoded in the learned representation. In addition, each POI's categorical label is embedded into a dense semantic vector $\mathbf{c}_i$ through another MLP, enabling the model to retain class-specific semantic information. The BERT-based contextual embedding and the category embedding are then fused to form a unified vector representation $\mathbf{e}_b^{\text{POI}}$ for each POI.

In parallel, we employ a pre-trained CLIP [39] text encoder to extract features from both the image caption and its corresponding long-form semantic description (Fig. 5). Each sentence is individually encoded, and their embeddings $\mathbf{e}_b^{\text{text}}$ are aggregated to construct a robust textual representation of the scene. Finally, the aggregated POI embedding and the textual representation are concatenated to form the Textual Knowledge:

$$\mathbf{t}_b = \text{Concat}(\mathbf{e}_b^{\text{POI}}, \mathbf{e}_b^{\text{text}}). \quad (7)$$

This fused vector encodes both structured spatial knowledge and human-centric semantic understanding, serving as a strong prior for downstream modules.

4.1.2 Visual Knowledge Extraction. This module is designed to capture both spatial and semantic cues from scene inputs, primarily focusing on aligned images and 3D point cloud. Specifically, for satellite images, we directly input them into the CLIP image encoder to obtain rich visual representations $\mathbf{v}_b^{\text{img}}$. For point cloud data, we project the 3D points onto a 2D plane along the Z-axis, generating an overhead view that preserves the geometric structure of the scene. This projected view is then processed by the same CLIP image encoder to capture structural-semantic features and spatial patterns $\mathbf{v}_b^{\text{pc}}$.

The visual features from satellite imagery and point cloud are complementary in nature. To effectively integrate these modalities, we apply the following Visual Transformation Module to fuse those features. Specifically, we apply average pooling and max pooling to the point cloud features, followed by an MLP layer with normalization and ReLU activation to transform them into a spatial weight distribution $\mathbf{w}_b \in \mathbb{R}^{H \times W}$:

$$\mathbf{v}_b = \mathbf{w}_b \odot \mathbf{v}_b^{\text{img}}. \quad (8)$$

This weight map is then multiplied with the image features on a per-pixel basis. By leveraging the global distribution from the complementary modality, we provide a prior that enhances the contextual understanding of visual knowledge.

4.2 Multimodal Contrastive Learning

To align textual and visual modalities within a unified geospatial representation space, we adopt a multimodal contrastive learning strategy inspired by CLIP. This approach facilitates cross-modality understanding by maximizing the agreement between paired textual and visual knowledge. The textual knowledge vector $\mathbf{t}_i$ and the visual knowledge vector $\mathbf{v}_i$ are first projected into a shared embedding space $\mathbf{t}'_i = \text{Proj}_t(\mathbf{t}_i)$, $\mathbf{v}'_i = \text{Proj}_v(\mathbf{v}_i)$ using separate linear projection heads. We then compute the cosine similarity $s_{i,j} = (\langle \mathbf{v}'_i, \mathbf{t}'_j \rangle) / (\|\mathbf{v}'_i\| \cdot \|\mathbf{t}'_j\|)$ between all pairs of text and visual embeddings within the batch. The contrastive loss is defined using a symmetric cross-entropy formulation:

$$\mathcal{L}_C = \frac{1}{2N} \sum_{i=1}^N \left[-\log \frac{\exp(s_{i,i}/\tau)}{\sum_{j=1}^N \exp(s_{i,j}/\tau)} - \log \frac{\exp(s_{i,i}/\tau)}{\sum_{j=1}^N \exp(s_{j,i}/\tau)} \right], \quad (9)$$

where N denotes the number of scene samples and τ is a learnable temperature parameter. Through this contrastive training objective, CLIP4Geo is encouraged to learn modality-invariant representations that retain both visual and geospatial semantics.

To enhance the consistency within each modality, we apply cosine similarity-based contrastive learning among features of the same modality. For textual features, we compute cosine similarities between the POI embeddings and the textual embeddings to ensure they represent consistent semantic information. Similarly, for visual modality, we calculate cosine similarity between features extracted from images and projected point clouds. The intra-modal contrastive loss encourages alignment within the same modality:

$$\mathcal{L}_{\text{intra}} = 1 - \cos(\mathbf{f}_1, \mathbf{f}_2), \quad (10)$$

where $\mathbf{f}_1$ and $\mathbf{f}_2$ are feature embeddings from the same modality.

4.3 Geospatial Context Awareness Transformer

To effectively build the geospatial awareness upon the alignment foundation, we propose the Geospatial Context Awareness Transformer as the core module of our framework. This module is specifically designed to capture spatial context and interactions across different modalities. It consists of Context-Aware Self Attention (C-A SA), Geospatial Cross Attention (G CA), and Modality Specific Feed-Forward Network (MS-FNN) (Fig. 3).

4.3.1 Context-Aware Self Attention. Standard self attention aggregates information globally, which may dilute critical local relationships. To address this limitation, we propose a context-aware self attention mechanism. We compute standard self attention separately for the textual and visual knowledge vectors, and incorporate the position embeddings $\mathbf{p}_i$ extracted from POIs as prior into both textual and visual knowledge vectors. Contextual information is then aggregated from the spatial neighbors associated with each position and added to the corresponding token representation to enhance local sensitivity.

4.3.2 Geospatial Cross Attention. To enhance cross-modality interaction, we insert cross attention modules between layers of the Transformer. These modules enable the knowledge vector from one modality to attend to another. First, we normalize the knowledge vectors obtained from the Context-Aware Self Attention module.

Location	Coverage		Image–LiDAR–Text pairs	Resolution	Pt Density	POI	Text Length
	Bottom-Left	Top-Right					
Austin (USA)	97.86°W, 30.10°N	97.58°W, 30.53°N	3,500	0.3m	> 12 Pts/m ²	5,000	213.8
California (USA)	122.10°W, 37.20°N	121.70°W, 37.50°N	10,800	1m	> 50 Pts/m ²	6,000	218.1
Zhangzhou (China)	24.32°E, 117.87°N	24.49°E, 118.19°N	400	1m	> 10 Pts/m ²	6,000	252.5

Table 2: Dataset benchmark description of the *CityVerse* benchmark across three urban regions.

Subsequently, we compute cross attention between the two modalities independently. Notably, since the knowledge embeddings are already spatially aligned and enriched with geospatial awareness through contextual position encoding, there is no need to introduce additional position embeddings at this stage. This design allows each knowledge vector to integrate token-level semantic relationships, its own geospatial context, and cross-modality geospatial context.

4.3.3 Modality Specific Feed-Forward Network. To accommodate the diversity of the two knowledge modalities, we adopt a modality-specific Feed-Forward Networks (MS-FFNs) design. After the cross attention, the output tokens are routed to distinct FFN sub-networks FFN_{ms} according to their modality. Each FFN sub-network possesses its own set of parameters and activation patterns, enabling modality-specific learning.

Note that not all FFN layers are modality-specific. In the lower layers, we use normal FFNs and share their parameters across all modalities to promote joint spatial reasoning and early semantic alignment. This shared design facilitates the learning of generalizable spatial relationships across modalities during the initial stages. In contrast, in the final two layers, we employ MS-FFNs to strengthen the representation of modality-dependent features and enhance the discriminative features of the final embeddings. This is motivated by the observation that, in higher layers, modalities tend to diverge semantically [48]. Hence, separate sub-networks are essential for preserving fine-grained, modality-specific information and preventing representational interference or degradation.

4.4 Multi-task Fine-tuning

To fully exploit the cross-modality representations learned from geospatial data, we formulate the training process as a multi-task learning problem. All tasks share a unified backbone to encourage knowledge transfer, while task-specific output heads are employed to retain task-specific sensitivity.

To further improve feature disentanglement and mitigate negative transfer between tasks, we perform task-specific layer normalization on MS-FFNs' output within each head. To avoid task imbalance and ensure stable convergence during training, we apply a dynamic task balancing strategy that adaptively reweights the loss contributions from each task based on their learning dynamics. Specifically, we monitor the training loss rate for each task and adjust their weights such that tasks with slower convergence receive higher gradients, following the principle of GradNorm [?] (or other method, if applicable). This allows all tasks to contribute meaningfully to shared backbone updates while reducing the risk of negative transfer. Furthermore, we define task-specific losses for each downstream task $\mathcal{L}_{task}$.

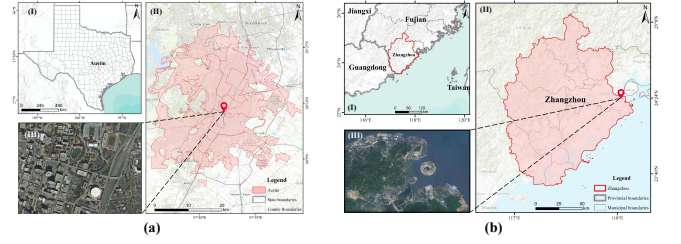


Figure 4: Data collection area of *CityVerse* Benchmark. (a) Urban region of Austin, Texas, USA, and (b) Urban region of Zhangzhou, Fujian, China. Note that the California region can be referred to [45]. The geographically and structurally diverse scenes from these regions support generalizable multimodal geospatial learning.

5 CityVerse Benchmark

In this section, we introduce our proposed large-scale benchmark *CityVerse*, which can be encoded as city-scale multimodal geospatial representation. To our knowledge, *CityVerse* is the first dataset that systematically integrates satellite imagery, LiDAR point clouds, POIs, and natural language description at the city block level.

5.1 Data Collection & Description

As illustrated in Fig. 4, the *CityVerse* Benchmark is built on a comprehensive collection of multimodal urban data with different urban form, function, and semantic complexity. Specifically, we focus on three representative areas: Austin (USA), California (USA) and Zhangzhou (China), to ensure geographic diversity and support cross-cultural urban analysis. Data collection involves four core modalities: satellite imagery, LiDAR point clouds, POIs, and scene text description. Notably, the satellite imagery and LiDAR point clouds of California are provided by the reference [45]. The information of the three datasets is shown in Tab. 2. Compared to existing benchmarks in Tab. 1, *CityVerse* uniquely integrates satellite imagery, LiDAR point clouds, POIs, and scene-level textual description at the urban-block scale, enabling comprehensive multimodal geospatial learning and supporting a broader range of downstream tasks.

Satellite Imagery. We collect high-resolution satellite imagery of Austin from the Inria Aerial Image Dataset [31], with a spatial resolution of 0.3 meters per pixel. The satellite imagery of Zhangzhou is obtained from the GF-2 satellite. Both the Zhangzhou and California imagery have a spatial resolution of 1 meter per pixel.

LiDAR Point Clouds. The LiDAR point cloud of Austin is obtained from airborne laser scanning data provided by the Texas Geographic

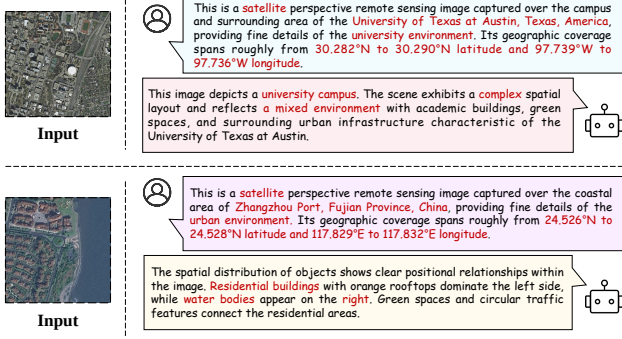


Figure 5: Examples of multimodal alignment in the CityVerse dataset. Each block is associated with one remote sensing image and two levels of textual description: (top) human-crafted prompts, and (bottom) long-form paragraph elaborating on spatial layout, object distribution, and semantic content. The examples from Austin (USA) and Zhangzhou (China) highlight the geospatial and semantic richness across regions.

Information Office¹. Meanwhile, we independently collect the point cloud of Zhangzhou, which features uniform density and includes per-point attributes such as RGB and intensity. The California point cloud is sourced from National Agriculture Imagery Program² and 3D Elevation Program³ Combined dataset.

Points of Interest (POIs). For Austin and California, POIs are extracted from OpenStreetMap (OSM)⁴, while POIs in Zhangzhou are obtained using Amap’s API⁵ due to its superior coverage and more detailed semantic tagging for Chinese cities. All POI records are converted to the WGS84 coordinate system and manually curated by geospatial experts to correct missing labels, misaligned coordinates, and ambiguous category assignments. Each POI is ultimately encoded as a tuple (x, y, c) , representing its spatial coordinates and categorical type.

Textual Scene Description. To construct scene-level semantic summaries, we collect textual description via two strategies: (1) rule-based caption generation using templates over POI and spatial attributes (Fig. 5) and (2) manual annotation from domain experts for selected urban blocks. Each block is paired with a short caption and an extended paragraph-level description. All text is normalized and tokenized using BERT-based preprocessing.

5.2 Data Statistics

We provide statistical and linguistic insights into the CityVerse dataset, as shown in Fig. 6.

Text Keyword Statistics. Fig. 6(a) presents a frequency histogram of keywords extracted from both captions and long-form description in Austin, California, and Zhangzhou. High-frequency terms include *residential*, *buildings*, *green*, *commercial*, *infrastructure*, and *vegetation*, indicating the dataset’s rich coverage of typical urban

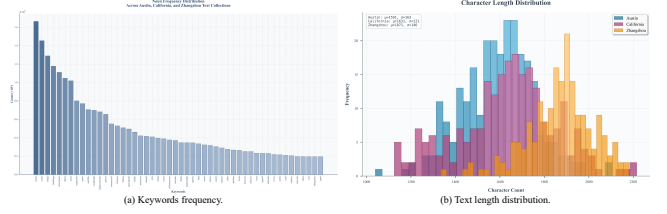


Figure 6: Statistical characteristics of the CityVerse dataset. (a) Caption keyword frequency, and (b) Caption length distribution.

functions and landscape types. Notably, keywords such as *satellite* and *high resolution* also appear frequently, reflecting the impact of geospatial sources and annotation styles across regions.

Text Length Distribution. Fig. 6(b) presents the distribution of caption lengths across the three regions in the CityVerse dataset. The average word counts are approximately 204 words for Austin, 218 for California, and 226 for Zhangzhou, with a long-tailed distribution in each region. While most captions fall within the 160–260 word range, a small number of entries exceed 280 words, especially in manually annotated description. This indicates high semantic richness and variability in textual content across regions.

6 Experiments

6.1 Settings & Configurations

We use AdamW optimizer with hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, batch size of 200, and a one-cycle cosine learning rate scheduler with a maximum learning rate of $5e-4$. The model is trained for 50 epochs on a single RTX A6000 48GB GPU. We utilize OpenCLIP implementation and fine-tune models on CityVerse datasets with parameter-efficient fine-tuning strategies, including Pfeiffer adapters and LoRA adapters.

We evaluate the CLIP4Geo tuned by the CityVerse dataset based on four downstream tasks: zero-shot classification, semantic segmentation / localization, cross-modality retrieval, and cross-modality generation. For zero-shot classification, we select the AID [47], RESISC45 [10], EuroSAT [18], and our CityVerse datasets. The evaluation metric is Top-1 accuracy. Moreover, we use our CityVerse to evaluate the semantic segmentation, and select AIR-SLT [56] dataset for semantic localization. We report Overall Accuracy (OA), mean Intersection over Union (mIoU), and four region-based metrics: R_{su} , R_{as} , R_{da} , and R_{mi} , respectively. For cross-modality retrieval, RSICDC [30] and RSITMD [55] datasets are utilized for performance comparison. We report Recall@K (K=1,5,10) for both text-to-image and image-to-text retrieval scenarios. Finally, we evaluate the performance of cross-modality generation using the CityVerse dataset. Specifically, BLEU [37] is used to assess the quality of short-form caption generation, while SPICE [3] is employed to evaluate paragraph-level semantic description. In addition, we use PSNR [15] to evaluate the quality of reconstructed images from masked inputs.

In addition, we propose a new metric - **Geographic Description Quality (GDQ)** - to evaluate the geographic correctness of generated text, particularly regarding the entities and spatial relationships. It consists of two components: **Type Accuracy (TA)**, which measures whether the geographic entities mentioned in the

¹<https://geographic.texas.gov>

²<https://naip-usdaonline.hub.arcgis.com/>

³<https://www.usgs.gov/3d-elevation-program>

⁴<https://www.openstreetmap.org/>

⁵<https://mobile.amap.com/>

Method	AID	RESISC45	EuroSAT	CityVerse
CLIP [39]	66.22	60.90	47.21	49.83
MobileCLIP [42]	63.50	58.11	44.20	45.34
BLIP [26]	68.14	63.95	51.84	54.07
BLIP2 [25]	72.08	70.03	61.23	62.56
SigLIP [59]	70.61	66.44	59.17	61.82
RSTeller [14]	69.65	70.98	52.56	-
Alpha-CLIP [40]	64.87	59.36	46.08	47.23
RemoteCLIP [28]	-	53.24	17.19	-
LRSClip [8]	-	73.12	57.55	-
CLIP4Geo	75.73	73.19	64.46	69.22

Table 3: Results for zero-shot classification task on AID, RESISC45, EuroSAT, and our proposed CityVerse datasets. Results are reported as Top-1 accuracy (%), and the best performance for each dataset is highlighted in bold.

Method	Segmentation		Localization			
	OA	mIoU	R _{su}	R _{as}	R _{da}	R _{mi}
SeLo-v1 [56]	-	-	0.69	0.33	0.67	0.68
SeLo-v2 [53]	-	-	0.72	0.29	0.67	0.70
CLIP [39]	92.85	87.56	0.72	0.30	0.70	0.71
MobileCLIP [42]	91.20	85.02	0.65	0.36	0.63	0.66
BLIP [26]	93.95	89.12	0.72	0.30	0.69	0.71
BLIP2 [25]	95.23	91.34	0.74	0.33	0.71	0.73
SigLIP [59]	94.57	90.80	0.73	0.27	0.70	0.72
Alpha-CLIP [40]	92.03	86.33	0.66	0.34	0.65	0.67
GeoRSCLIP [62]	-	-	0.73	0.30	0.69	0.71
CLIP4Geo	99.08	94.69	0.76	0.26	0.73	0.74

Table 4: Results for semantic segmentation / localization tasks on our CityVerse and AIR-SLT, respectively. Segmentation metrics are reported as percentages (%), while localization metrics are reported as decimal fractions. The best results are highlighted in bold.

text match the actual POI types; and **Relation Accuracy (RA)**, which assesses whether the described spatial relationships (e.g., relative positions of identified objects in the image, such as at the left corner) align with the real-world spatial layout. The final GDQ score is computed as a weighted average of the two:

$$\text{GDQ} = \lambda_1 \cdot \text{TA} + \lambda_2 \cdot \text{RA}, \quad \text{where } \lambda_1 + \lambda_2 = 1, \quad (11)$$

In our experiments, we set $\lambda_1 = 0.6$ and $\lambda_2 = 0.4$ to emphasize the accuracy of geographic entity recognition. We employ four representative large language models (LLMs; i.e., GPT-4o [34], DeepSeek-VL [29], Claude 3 [4], and Gemini Pro [41]) to evaluate the GDQ.

6.2 Downstream Application Results

To evaluate the generalization ability of CLIP4Geo and the usefulness and CityVerse, we conduct comprehensive downstream experiments across four representative geospatial tasks as below.

6.2.1 Zero-shot Classification. Tab. 3 summarizes the zero-shot classification results on four benchmark datasets: AID, RESISC45, EuroSAT, and CityVerse. CLIP4Geo achieves the highest Top-1

Method	I2T Retri.			T2I Retri.		
	R@1	R@5	R@10	R@1	R@5	R@10
RSICD						
LW-MCR [57]	3.29	12.52	19.93	4.66	17.51	30.02
VSE++	3.38	9.51	17.46	2.82	11.32	18.10
AFMFN [55]	5.39	15.08	23.40	4.90	18.28	31.44
GaLR [58]	6.59	19.85	31.04	4.69	19.48	32.13
SWAN [36]	7.41	20.13	30.86	5.56	22.26	37.41
HVSA	7.47	20.62	32.11	5.51	21.13	34.13
PIR [35]	9.88	27.26	39.16	6.97	24.56	38.92
FAAMI	10.44	22.66	30.89	8.11	25.59	41.37
PE-RSITR [54]	14.13	31.51	44.78	11.63	33.92	50.71
MTGFE	15.28	37.05	51.60	8.67	27.56	43.92
CLIP [39]	5.31	14.18	23.70	5.78	17.73	27.76
MSACLIP [51]	7.87	25.67	41.85	10.16	25.71	36.96
RemoteCLIP [28]	17.02	37.97	51.51	13.71	31.11	54.25
CLGSA [9]	11.75	34.07	50.12	10.72	32.30	45.57
MobileCLIP [42]	5.33	14.10	23.69	5.79	17.65	27.70
BLIP [26]	9.12	23.10	35.15	9.34	22.25	34.20
BLIP2 [25]	10.60	26.15	39.00	10.95	25.50	37.85
SigLIP [59]	9.62	24.31	36.95	9.83	23.75	35.65
Alpha-CLIP [40]	9.75	24.70	37.25	9.95	24.20	36.08
MLT [2]	10.70	29.64	41.53	9.14	28.96	44.59
LRSClip [8]	10.01	28.60	43.73	13.08	31.38	41.26
CLIP4Geo	21.54	40.71	56.89	14.12	36.38	52.46
RSITMD						
LW-MCR [57]	10.18	28.98	39.82	7.79	30.18	49.78
VSE++	10.38	27.65	39.60	7.79	24.87	38.67
AFMFN [55]	11.06	29.20	38.72	9.96	34.03	52.96
HVSA	13.20	32.08	45.58	11.43	39.20	57.45
SWAN [36]	13.35	32.15	46.90	11.24	40.20	60.60
GaLR [58]	14.82	31.64	42.48	11.16	36.68	51.65
FAAMI	16.15	35.62	48.89	12.96	42.39	59.96
MTGFE	17.92	40.93	53.32	16.59	48.50	67.43
PIR [35]	18.14	41.15	52.88	12.17	41.68	63.41
PE-RSITR [54]	23.67	44.07	60.36	20.10	50.63	67.97
CLIP [39]	9.51	23.01	32.74	8.81	27.88	43.19
RemoteCLIP [28]	27.88	50.66	65.71	22.17	36.36	73.41
MobileCLIP [42]	8.96	20.45	39.52	11.62	26.50	49.71
BLIP [26]	16.85	38.90	58.25	15.88	33.60	61.55
BLIP2 [25]	17.32	40.05	59.93	16.22	34.15	62.34
SigLIP [59]	16.92	29.30	38.73	15.91	33.72	51.92
Alpha-CLIP [40]	16.88	29.10	38.62	15.86	33.58	51.80
MLT [2]	19.69	40.26	54.42	17.61	49.73	66.59
LRSClip [8]	17.52	41.68	56.73	21.02	39.38	52.21
CLIP4Geo	28.35	51.10	66.10	22.05	37.10	74.03

Table 5: Performance comparison on RSICD and RSITMD datasets for image-text (I2T) and text-image (T2I) retrievals, with reported Recall @K, where K = 1, 5, 10. Bold numbers highlight the best results in each column.

accuracy on all datasets, significantly outperforming the general-purpose CLIP (e.g., 75.73% vs. 66.22% on AID and 69.22% vs. 49.83% on CityVerse). The improvements are especially pronounced on CityVerse, suggesting that the integration of structured geospatial context and POI information enhances model performance in complex real-world scenes. Compared to strong baselines, CLIP4Geo achieves 5–10% gains, demonstrating superior category alignment in unseen geospatial scenes.

Method	Cap. / Desc. Gen.		Mask Image Recons.
	BLEU	SPICE	PSNR
CLIPCap [33]	16.42	18.91	23.15
BLIP [26]	18.85	20.32	24.06
BLIP2 [25]	22.13	21.95	24.85
CLIP4Geo	24.87	23.68	26.43

Table 6: Results for cross-modality generation task on our CityVerse dataset. The best results are in bold.

Method	GPT-4o	DeepSeek-VL	Claude 3	Gemini Pro
CLIPCap [33]	0.28	0.18	0.20	0.15
BLIP [26]	0.35	0.30	0.32	0.29
BLIP2 [25]	0.32	0.66	0.58	0.65
CLIP4Geo	0.63	0.88	0.90	0.93

Table 7: Geographic Description Quality (GDQ) evaluation across foundation models on the CityVerse Dataset. The best results are in bold.

6.2.2 Semantic Segmentation / Localization. In Tab. 4, CLIP4Geo delivers state-of-the-art performance in both pixel-level segmentation and region-based semantic localization. On the CityVerse dataset, it achieves an overall accuracy of 99.08% and a mIoU of 94.69%, exceeding all baselines by a clear margin. On the AIR-SLT localization benchmark, CLIP4Geo also achieves the best scores on three out of four regional metrics, with notable improvements in region similarity ($R_{su} = 0.76$) and mean region consistency ($R_{mi} = 0.74$). These results highlight the benefit of geospatial context awareness for modeling region boundaries and complex spatial relationships.

6.2.3 Cross-modality Retrieval. We evaluate cross-modality retrieval using RSICD and RSITMD datasets for image-to-text (I2T) and text-to-image (T2I) retrieval. As shown in Tab. 5, CLIP4Geo consistently surpasses all baselines in both directions, achieving a Recall@1 of 28.35% (I2T) and 22.05% (T2I) on RSICD, and 21.54% (I2T) and 14.12% (T2I) on RSITMD. Notably, it outperforms RemoteCLIP and PE-RSITR by substantial margins (e.g., +5–7% in Recall@10). The integration of LiDAR, POI, and text information into a unified geospatial representation significantly improves semantic alignment between modalities, especially in scenes with dense object distribution or semantic ambiguity.

6.2.4 Cross-modality Generation. CLIP4Geo exhibits strong generative capabilities for both short-form captioning and long-form description synthesis. As shown in Tab. 6, it achieves the highest BLEU (24.87) and SPICE (23.68) scores among all compared models, indicating that it generates text that is both linguistically fluent and semantically accurate. Moreover, CLIP4Geo attains the best PSNR (26.43) in the masked image reconstruction task, demonstrating its capacity to capture fine-grained structural information across modalities.

To further evaluate geographic fidelity, as reported in Tab. 7, CLIP4Geo achieves the highest GDQ across all LLM evaluators, with scores up to 0.93 (Gemini Pro) and 0.90 (Claude 3), significantly higher than CLIPCap and BLIP-based models. This confirms the effectiveness of incorporating geospatial priors and alignment

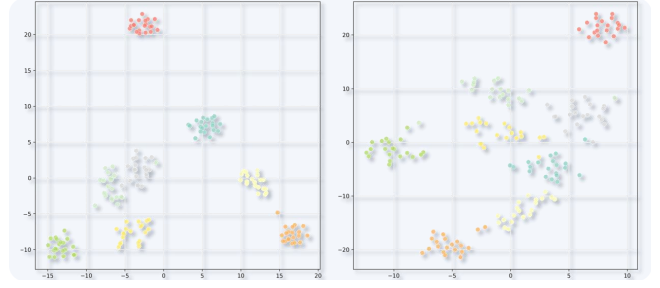


Figure 7: Ablation study of cross-modality alignment: CLIP4Geo (left) vs. Baseline (right)

mechanisms, especially for entity-type recognition and spatial relation understanding.

6.3 Ablation Studies for RQ1

Fig. 2 demonstrates the impact of geospatial context awareness on the model’s attention behavior. Without incorporating geospatial context from POIs, the attention heatmap appears diffuse and lacks semantic focus. In contrast, when geospatial context is integrated, the attention becomes more concentrated on semantically meaningful regions, particularly around POI-located areas. These results validate that geospatial alignment enhances the model’s ability to capture cross-modality spatial dependencies and improves its focus on informative regions, which is crucial for downstream tasks such as caption generation and cross-modality retrieval.

To evaluate whether incorporating geospatial context improves the semantic structure and discriminative quality of learned representations, we visualize the feature embeddings using t-SNE. As shown in Fig. 7, CLIP4Geo produces well-separated and compact clusters in the left sub-figure, in contrast to the baseline CLIP (right sub-figure), which exhibits more dispersed and overlapping distributions. Quantitatively, our approach achieves a silhouette score of 0.764, significantly higher than the baseline’s 0.653. This indicates that geospatial context can help GeoFMs obtain representations with stronger intra-class coherence and inter-class separation.

6.4 Ablation Studies for RQ2

Fig. 8(a) visualizes the pairwise cosine similarity between image, text, POI, and point cloud embeddings across five different city blocks. The clear diagonal blocks reflect consistent intra-scene alignment, indicating that our model effectively fuses diverse geospatial cues. In Fig. 8(b), we ablate $\mathcal{L}_{intra}$ and $\mathcal{L}_C$ to examine their roles in retrieval accuracy. Removing either component causes notable degradation, while the full model achieves the best results across all recall metrics. These findings confirm the necessity of multi-level alignment objectives (i.e., intra- and cross-modality) in learning and bridging the semantic gap between different modalities.

In Fig. 9(a), embeddings from different modalities within the same scene are consistently clustered together, revealing effective cross-modality alignment. Furthermore, Fig. 9(b) displays that our CLIP4Geo can accurately retrieve the ground truth among similar samples. These results validate the effectiveness of our multi-level alignment strategy and provide a feasible solution for RQ2.

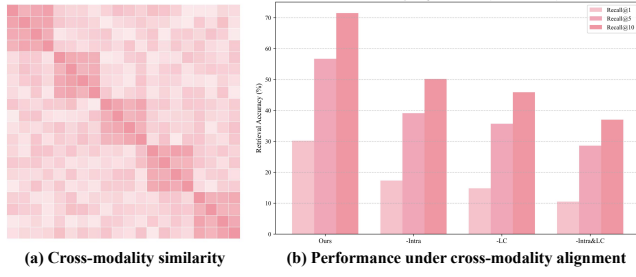


Figure 8: Ablation study of cross-modality alignment

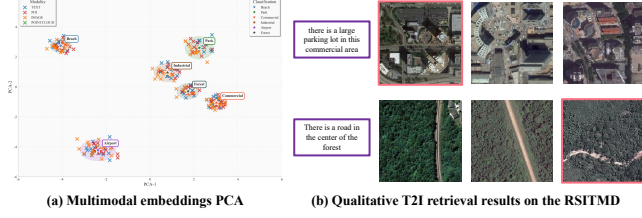


Figure 9: (a) PCA projection of multimodal embeddings. It presents six scene categories, each represented by an ellipse with semantic scene points. Each point denotes a modality-specific embedding. (b) Qualitative T2I retrieval results on the RSITMD dataset. The ground-truth images and texts are marked with pink and purple, respectively.

Method	Fine-Tune	I2T R@10	T2I R@10
RemoteCLIP	RSITMD	55.23	62.41
RemoteCLIP	RET-3/10 + SEG-4	65.71	73.41
RemoteCLIP	RS5M	61.02	71.35
RemoteCLIP	<i>CityVerse</i>	64.85	73.12
GeoRSCLIP	RSITMD	42.83	54.08
GeoRSCLIP	RS5M	46.46	57.52
GeoRSCLIP	RS5M + RSITMD	63.27	74.60
GeoRSCLIP	RS5M + RET-2	67.92	74.38
GeoRSCLIP	<i>CityVerse</i>	65.97	73.88

Table 8: Cross-modality retrieval results of different models fine-tuned on our CityVerse and other datasets. The best results are in bold.

6.5 Ablation Studies for RQ3

Tab. 8 presents the cross-modality retrieval performance of RemoteCLIP and GeoRSCLIP models fine-tuned on various datasets. Notably, RemoteCLIP and GeoRSCLIP fine-tuned on our proposed CityVerse dataset achieve strong results. Despite containing significantly fewer samples than RS5M or RET-3/10 + SEG-4, CityVerse offers structured geographic knowledge that enhances cross-modality alignment and geospatial reasoning. These results highlight that our well-curated and domain-specific dataset CityVerse is able to provide competitive or even superior generalization capabilities compared to massive-scale but weakly annotated corpora.



Figure 10: Failure case visualization. Due to the semantic inconsistency (coarse POI taxonomy and POI sparsity) between the POI class and the generated textual description, the model’s attention is incorrectly focused on non-commercial buildings, rather than the actual commercial target.

7 Discussion

7.1 Innovation Summary and RQ Reflection

First, we address RQ1 by explicitly incorporating POI categories, scene-level description, geographic coordinates, satellite images, and LiDAR to enhance the model’s geospatial context awareness. Second, we contribute directly to RQ2 by introducing an effective alignment strategy, which improves the cross-modality discriminability and intra-modality coherence of the embeddings. Moreover, we construct CityVerse, a high-quality multimodal geospatial dataset, to provide a semantically rich benchmark for developing GeoFMs, thereby answering RQ3 regarding the importance of dataset quality over scale.

7.2 Limitations and Future Work

While CityVerse supports diverse urban forms, its geographic scope is still limited. Future work could integrate global satellite-ground paired modalities to improve model generalization to unseen geographies. Moreover, the quality of POI data has a significant impact on the effectiveness of expert knowledge systems, as shown in Fig. 10. Also, the POI taxonomy used is relatively coarse. In future work, we plan to detect and address semantic inconsistency between POIs and other modalities, introduce finer-grained POI categories and functional hierarchies, and leverage hyperbolic space learning to further enrich geospatial prompts.

8 Conclusion

This work proposes a unified GeoFM, *CLIP4Geo*, which effectively integrates satellite imagery, LiDAR point clouds, geo-tagged POIs, expert-guided textual description, to support downstream geospatial tasks. Our approach introduces a Geospatial Expert Knowledge System to encode tailored, multimodal data into shared geospatial representation. Furthermore, we design Geospatial Context Awareness Transformer that employs geospatial context position embedding to guide the cross-attention mechanism, which effectively captures spatial dependencies across modalities. We also curate *CityVerse* to provide a high-quality multimodal geospatial benchmark for model training and evaluation. Extensive experiments demonstrate that our model significantly improves representation discriminability and downstream performance. Ablation studies and visualization analyses further verify the necessity of incorporating geospatial context and multimodal alignment in various geospatial tasks.

Acknowledgment

This work was supported by the 2025 University of Glasgow Early Career Reward for Excellence on "Developing Vision–Language Models Enhanced by Geospatial Intelligence" and the 2025 University of Glasgow Early Career Mobility Scheme (ECMS) Award.

References

- [1] Mohit Agarwal, Mimi Sun, Chaitanya Kamath, Arbaaz Muslim, Prithul Sarker, Joydeep Paul, Hector Yee, Marcin Sieniek, Kim Jablonski, Yael Mayer, et al. 2024. General Geospatial Inference with a Population Dynamics Foundation Model. *arXiv preprint arXiv:2411.07207* (2024).
- [2] Mohamad M Al Rahhal, Yakoub Bazi, Norah A Alsharif, Laila Bashmal, Naif Alajlan, and Farid Melgani. 2022. Multilanguage transformer for improved text to remote sensing image retrieval. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 15 (2022), 9115–9126.
- [3] Peter Anderson, Basura Fernando, Mark Johnson, and Stephen Gould. 2016. Spice: Semantic propositional image caption evaluation. In *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part V 14*. Springer, 382–398.
- [4] Anthropic. 2024. Introducing Claude 3.5 Sonnet. <https://www.anthropic.com/news/claude-3-7-sonnet?guides=image-generation-social-good>
- [5] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. 2023. Qwen-vl: A frontier large vision-language model with versatile abilities. *arXiv preprint arXiv:2308.12966* 1, 2 (2023), 3.
- [6] Prabin Bhandari, Antonios Anastasopoulos, and Dieter Pfoser. 2024. Urban mobility assessment using llms. In *Proceedings of the 32nd ACM International Conference on Advances in Geographic Information Systems*. 67–79.
- [7] Nikolaos Ioannis Bountos, Arthur Ouaknine, Ioannis Papoutsis, and David Rolnick. 2025. Fomo: Multi-modal, multi-scale and multi-task remote sensing foundation models for forest monitoring. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 27858–27868.
- [8] Weizhi Chen, Jingbo Chen, Yupeng Deng, Jiansheng Chen, Yuman Feng, Zhihao Xi, Diyou Liu, Kai Li, and Yu Meng. 2025. LRSCLIP: A Vision-Language Foundation Model for Aligning Remote Sensing Image with Longer Text. *arXiv preprint arXiv:2503.19311* (2025).
- [9] Xiumei Chen, Xiangtao Zheng, and Xiaoqiang Lu. 2025. Context-Aware Local-Global Semantic Alignment for Remote Sensing Image-Text Retrieval. *IEEE Transactions on Geoscience and Remote Sensing* (2025).
- [10] Gong Cheng, Junwei Han, and Xiaoqiang Lu. 2017. Remote sensing image scene classification: Benchmark and state of the art. *Proc. IEEE* 105, 10 (2017), 1865–1883.
- [11] Yezhen Cong, Samar Khanna, Chenlin Meng, Patrick Liu, Erik Rozi, Yutong He, Marshall Burke, David Lobell, and Stefano Ermon. 2022. Satmae: Pre-training transformers for temporal and multi-spectral satellite imagery. *Advances in Neural Information Processing Systems* 35 (2022), 197–211.
- [12] Florinel-Alin Croitoru, Vlad Hondru, Radu Tudor Ionescu, and Mubarak Shah. 2023. Diffusion models in vision: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 9 (2023), 10850–10869.
- [13] Casper Fibæk, Luke Camilleri, Andreas Luyts, Nikolaos Dionelis, and Bertrand Le Saux. 2024. PhilEO bench: Evaluating geo-spatial foundation models. In *IGARSS 2024-2024 IEEE International Geoscience and Remote Sensing Symposium*. IEEE, 2739–2744.
- [14] Junyao Ge, Xu Zhang, Yang Zheng, Kaitai Guo, and Jimin Liang. 2025. Rsteller: Scaling up visual language modeling in remote sensing with rich linguistic semantics from openly available data and large language models. *ISPRS Journal of Photogrammetry and Remote Sensing* 226 (2025), 146–163.
- [15] Rafael C Gonzales and Paul Wintz. 1987. *Digital image processing*. Addison-Wesley Longman Publishing Co., Inc.
- [16] Xixuan Hao, Wei Chen, Yibo Yan, Siru Zhong, Kun Wang, Qingsong Wen, and Yuxuan Liang. 2025. UrbanVLP: Multi-granularity vision-language pretraining for urban socioeconomic indicator prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 28061–28069.
- [17] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. 2022. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 16000–16009.
- [18] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. 2019. Eurosat: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* 12, 7 (2019), 2217–2226.
- [19] Chia-Yu Hsu, Wenwen Li, and Sizhe Wang. 2024. Geospatial foundation models for image analysis: Evaluating and enhancing NASA-IBM Prithvi’s domain adaptability. *International Journal of Geographical Information Science* (2024), 1–30.
- [20] Jincai Huang, Yongjun Xu, Qi Wang, Qi Cheems Wang, Xingxing Liang, Fei Wang, Zhao Zhang, Wei Wei, Boxuan Zhang, Libo Huang, et al. 2025. Foundation models and intelligent decision-making: Progress, challenges, and perspectives. *The Innovation* (2025).
- [21] Pallavi Jain, Dino Ienco, Roberto Interdonato, Tristan Berchoux, and Diego Marcos. 2025. SenCLIP: Enhancing zero-shot land-use mapping for Sentinel-2 with ground-level prompting. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. IEEE, 5656–5665.
- [22] Yuhan Ji, Song Gao, Ying Nie, Ivan Majić, and Krzysztof Janowicz. 2025. Foundation models for geospatial reasoning: assessing the capabilities of large language models in understanding geometries and topological spatial relations. *International Journal of Geographical Information Science* (2025), 1–38.
- [23] Konstantin Klemmer, Esther Rolf, Caleb Robinson, Lester Mackey, and Marc Rußwurm. 2025. Satclip: Global, general-purpose location embeddings with satellite imagery. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 4347–4355.
- [24] Alexandre Lacoste, Nils Lehmann, Pau Rodriguez, Evan Sherwin, Hannah Kerner, Björn Lütjens, Jeremy Irvin, David Dao, Hamed Alemohammad, Alexandre Drouin, et al. 2023. Geo-bench: Toward foundation models for earth monitoring. *Advances in Neural Information Processing Systems* 36 (2023), 51080–51093.
- [25] Junnan Li, Dongxu Li, Silvio Savarese, and Steven Hoi. 2023. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*. PMLR, 19730–19742.
- [26] Junnan Li, Dongxu Li, Caiming Xiong, and Steven Hoi. 2022. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International conference on machine learning*. PMLR, 12888–12900.
- [27] Zijing Liang, Yanjie Xu, Yifan Hong, Penghui Shang, Qi Wang, Qiang Fu, and Ke Liu. 2024. A Survey of Multimodal Large Language Models. In *Proceedings of the 3rd International Conference on Computer, Artificial Intelligence and Control Engineering*. 405–409.
- [28] Fan Liu, Delong Chen, Zhangqingyun Guan, Xiacong Zhou, Jiale Zhu, Qiaolin Ye, Liyong Fu, and Jun Zhou. 2024. Remoteclip: A vision language foundation model for remote sensing. *IEEE Transactions on Geoscience and Remote Sensing* (2024).
- [29] Haoyu Lu, Wen Liu, Bo Zhang, Bingxuan Wang, Kai Dong, Bo Liu, Jingxiang Sun, Tongzheng Ren, Zhuoshu Li, Hao Yang, et al. 2024. Deepseek-vl: towards real-world vision-language understanding. *arXiv preprint arXiv:2403.05525* (2024).
- [30] Xiaoqiang Lu, Binqiang Wang, Xiangtao Zheng, and Xuelong Li. 2017. Exploring Models and Data for Remote Sensing Image Caption Generation. *IEEE Transactions on Geoscience and Remote Sensing* 56, 4 (2017), 2183–2195.
- [31] Emmanuel Maggiori, Yuliya Tarabalka, Guillaume Charpiat, and Pierre Alliez. 2017. Can Semantic Labeling Methods Generalize to Any City? The Inria Aerial Image Labeling Benchmark. In *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*. IEEE.
- [32] Li Mi, Xianjie Dai, Javiera Castillo-Navarro, and Devis Tuia. 2024. Knowledge-aware text-image retrieval for remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing* (2024).
- [33] Ron Mokady, Amir Hertz, and Amit H Bermano. 2021. Clipcap: Clip prefix for image captioning. *arXiv preprint arXiv:2111.09734* (2021).
- [34] OpenAI and et al. Josh Achiam. 2024. GPT-4 Technical Report. *arXiv:2303.08774 [cs.CL]* <https://arxiv.org/abs/2303.08774>
- [35] Jiancheng Pan, Muyuan Ma, Qing Ma, Cong Bai, and Shengyong Chen. 2024. Pir: Remote sensing image-text retrieval with prior instruction representation learning. *arXiv preprint arXiv:2405.10160* (2024).
- [36] Jiancheng Pan, Qing Ma, and Cong Bai. 2023. Reducing semantic confusion: Scene-aware aggregation network for remote sensing cross-modal retrieval. In *Proceedings of the 2023 ACM International Conference on Multimedia Retrieval*. 398–406.
- [37] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*. 311–318.
- [38] Yujia Qin, Shengding Hu, Yankai Lin, Weize Chen, Ning Ding, Ganqu Cui, Zheni Zeng, Xuanhe Zhou, Yufei Huang, Chaojun Xiao, et al. 2024. Tool learning with foundation models. *Comput. Surveys* 57, 4 (2024), 1–40.
- [39] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PMLR, 8748–8763.
- [40] Zeyi Sun, Ye Fang, Tong Wu, Pan Zhang, Yuhang Zang, Shu Kong, Yuanjun Xiong, Dahua Lin, and Jiaqi Wang. 2024. Alpha-clip: A clip model focusing on wherever you want. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 13019–13029.
- [41] Team, Gemini, Georgiev, et al. 2024. Gemini 1.5: Unlocking multimodal understanding across millions of tokens of context. *arXiv preprint arXiv:2403.05530* (2024).
- [42] Pavan Kumar Anasosalu Vasu, Hadi Pouransari, Fartash Faghri, Raviteja Vemulapalli, and Oncel Tuzel. 2024. Mobileclip: Fast image-text models through multi-modal reinforced training. In *Proceedings of the IEEE/CVF Conference on*

- Computer Vision and Pattern Recognition*. 15963–15974.
- [43] Vicente Vivanco Cepeda, Gaurav Kumar Nayak, and Mubarak Shah. 2023. Geoclip: Clip-inspired alignment between locations and images for effective worldwide geo-localization. *Advances in Neural Information Processing Systems* 36 (2023), 8690–8701.
 - [44] Xiao Wang and Guo-Jun Qi. 2022. Contrastive learning with stronger augmentations. *IEEE transactions on pattern analysis and machine intelligence* 45, 5 (2022), 5549–5560.
 - [45] Yameng Wang, Yi Wan, Yongjun Zhang, Bin Zhang, and Zhi Gao. 2023. Imbalance knowledge-driven multi-modal network for land-cover semantic segmentation using aerial images and LiDAR point clouds. *ISPRS Journal of Photogrammetry and Remote Sensing* 202 (2023), 385–404.
 - [46] Junjie Wu, Jiangtao Xie, Zhaolin Zhang, Qilong Wang, Qinghua Hu, Peihua Li, and Sen Xu. 2025. DALIP: Distribution Alignment-based Language-Image Pre-Training for Domain-Specific Data. *arXiv preprint arXiv:2504.01386* (2025).
 - [47] Gui-Song Xia, Jingwen Hu, Fan Hu, Baoguang Shi, Xiang Bai, Yanfei Zhong, Liangpei Zhang, and Xiaoqiang Lu. 2017. AID: A benchmark data set for performance evaluation of aerial scene classification. *IEEE Transactions on Geoscience and Remote Sensing* 55, 7 (2017), 3965–3981.
 - [48] Congxi Xiao, Jingbo Zhou, Yixiong Xiao, Jizhou Huang, and Hui Xiong. 2024. ReFound: Crafting a Foundation Model for Urban Region Understanding upon Language and Visual Foundations. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 3527–3538.
 - [49] Zhitong Xiong, Fahong Zhang, Yi Wang, Yilei Shi, and Xiao Xiang Zhu. 2024. EarthNets: Empowering artificial intelligence for Earth observation. *IEEE Geoscience and Remote Sensing Magazine* (2024).
 - [50] Yibo Yan, Haomin Wen, Siru Zhong, Wei Chen, Haodong Chen, Qingsong Wen, Roger Zimmermann, and Yuxuan Liang. 2024. Urbanclip: Learning text-enhanced urban region profiling with contrastive language-image pretraining from the web. In *Proceedings of the ACM Web Conference 2024*. 4006–4017.
 - [51] Rui Yang, Shuang Wang, Yingping Han, Yuanheng Li, Dong Zhao, Dou Quan, Yanhe Guo, Licheng Jiao, and Zhi Yang. 2024. Transcending fusion: A multi-scale alignment method for remote sensing image-text retrieval. *IEEE Transactions on Geoscience and Remote Sensing* (2024).
 - [52] Christopher Yeh, Chenlin Meng, Sherrie Wang, Anne Driscoll, Erik Rozi, Patrick Liu, Jihyeon Lee, Marshall Burke, David B Lobell, and Stefano Ermon. 2021. Sustainbench: Benchmarks for monitoring the sustainable development goals with machine learning. *arXiv preprint arXiv:2111.04724* (2021).
 - [53] Miao Yu, Heqiang Yuan, Jialiang Chen, Chongyang Hao, Zhe Wang, Zhiqiang Yuan, and Bin Lu. 2023. SeLo v2: Toward for higher and faster semantic localization. *IEEE Geoscience and Remote Sensing Letters* 20 (2023), 1–5.
 - [54] Yuan Yuan, Yang Zhan, and Zhitong Xiong. 2023. Parameter-efficient transfer learning for remote sensing image-text retrieval. *IEEE Transactions on Geoscience and Remote Sensing* 61 (2023), 1–14.
 - [55] Zhiqiang Yuan, Wenkai Zhang, Kun Fu, Xuan Li, Chubo Deng, Hongqi Wang, and Xian Sun. 2022. Exploring a fine-grained multiscale method for cross-modal remote sensing image retrieval. *arXiv preprint arXiv:2204.09868* (2022).
 - [56] Zhiqiang Yuan, Wenkai Zhang, Chongyang Li, Zhaoying Pan, Yongqiang Mao, Jialiang Chen, Shouke Li, Hongqi Wang, and Xian Sun. 2022. Learning to evaluate performance of multi-modal semantic localization. *arXiv preprint arXiv:2209.06515* (2022).
 - [57] Zhiqiang Yuan, Wenkai Zhang, Xuee Rong, Xuan Li, Jialiang Chen, Hongqi Wang, Kun Fu, and Xian Sun. 2021. A lightweight multi-scale crossmodal text-image retrieval method in remote sensing. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2021), 1–19.
 - [58] Zhiqiang Yuan, Wenkai Zhang, Changyuan Tian, Xuee Rong, Zhengyuan Zhang, Hongqi Wang, Kun Fu, and Xian Sun. 2022. Remote sensing cross-modal text-image retrieval based on global and local information. *IEEE Transactions on Geoscience and Remote Sensing* 60 (2022), 1–16.
 - [59] Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. Sig-moid loss for language image pre-training. In *Proceedings of the IEEE/CVF international conference on computer vision*. 11975–11986.
 - [60] Wei Zhang, Miaoxin Cai, Tong Zhang, Yin Zhuang, Jun Li, and Xuerui Mao. 2024. Earthmarker: A visual prompting multi-modal large language model for remote sensing. *IEEE Transactions on Geoscience and Remote Sensing* (2024).
 - [61] Yifan Zhang, Cheng Wei, Zhengting He, and Wenhao Yu. 2024. GeoGPT: An assistant for understanding and processing geospatial tasks. *International Journal of Applied Earth Observation and Geoinformation* 131 (2024), 103976.
 - [62] Zilun Zhang, Tiancheng Zhao, Yulong Guo, and Jianwei Yin. 2024. RS5M and GeoRSClip: A large scale vision-language dataset and a large vision-language model for remote sensing. *IEEE Transactions on Geoscience and Remote Sensing* (2024).
 - [63] Yuanxin Zhao, Mi Zhang, Bingnan Yang, Zhan Zhang, Jujia Kang, and Jianya Gong. 2025. LuoJiaHOG: A hierarchy oriented geo-aware image caption dataset for remote sensing image-text retrieval. *ISPRS Journal of Photogrammetry and Remote Sensing* 222 (2025), 130–151.
 - [64] Zuopeng Zhao, Xiaoran Miao, Lei Liu, Xinzhen Xu, Ying Liu, Jianfeng Hu, Bingbing Min, Yumeng Gao, and Kanyaphakphachorn Pharksuwan. 2025. Sparse-Guided Partial Dense for Cross-Modal Remote Sensing Image-Text Retrieval. *IEEE Transactions on Geoscience and Remote Sensing* (2025).